Simulation and Reconstruction Study of Quirk Signatures at the LHC

Reproduction study based on Knapen, Lou, Papucci, Setford, “Tracking down Quirks at the Large Hadron Collider,” arXiv:1708.02243v2

Project directory: /home/rafia/coplanarity_study | Report date: 2026-09-15

1. Purpose of the Study

This project rebuilds, checks, and where needed corrects a computer simulation of “quirks” — a hypothetical particle pair joined by a string-like force — as they would appear inside the ATLAS detector at the LHC. The goal is to reproduce the search method described in the reference paper: instead of fitting a normal curved track, the method looks for detector hits from both particles that lie on one flat plane. The study covers signal generation, particle motion, detector response, the plane-fitting algorithm, and the background processes that could fake a signal.

2. Complete Simulation Workflow

The pipeline has seven stages, run in order for every simulated event:

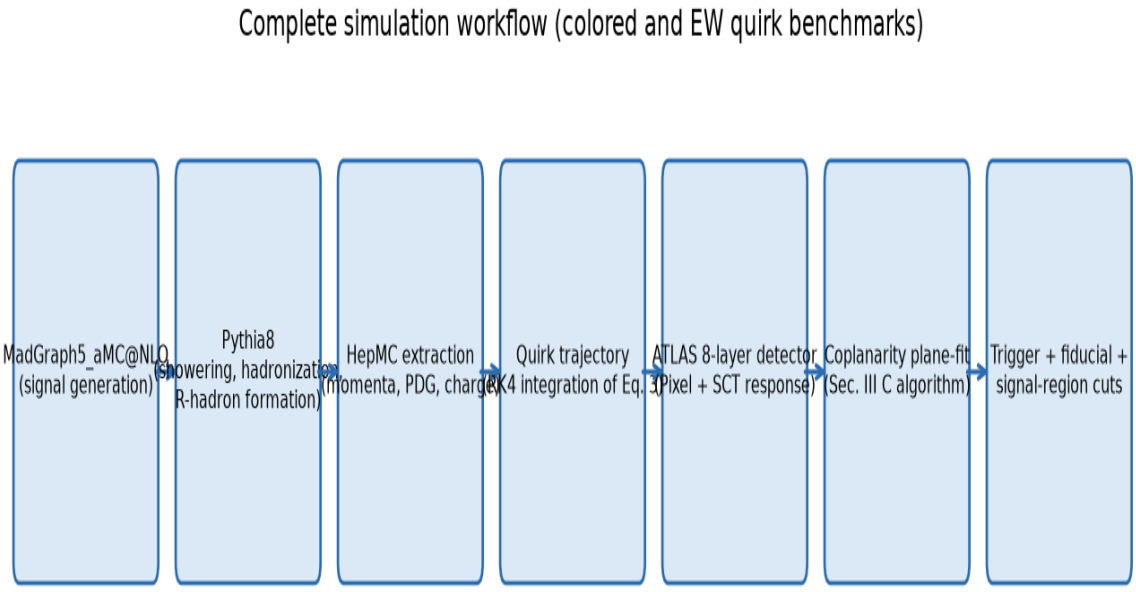


Figure 1. Complete simulation workflow, from event generation to the final signal-region cuts.

Interpretation:

- Each stage's output file feeds directly into the next stage — no manual steps in between.
- Stages 1–2 (MadGraph5_aMC@NLO + Pythia8) run outside this project; stages 3–7 are the code in /home/rafia/coplanarity_study.

3. Colored and Electroweak Benchmark Values

Two benchmark quirk types are simulated, matching the paper's own benchmark choices.

Quantity	Colored quirk	EW quirk	Paper ref.
Representation	(3,1)_{2/3}	(1,1)_1	Sec. II

Quantity	Colored quirk	EW quirk	Paper ref.
Quirk mass, m_Q	1800 GeV	800 GeV	Table I
String scale, Λ	2 keV	4 keV	Fig. 5
Electric charge	+2/3 (per-event, not all pairs usable)	+1 (qe-) / +1 (qe+)	Sec. II, III-A
Production mode	QCD	Drell-Yan	Sec. III-A
Magnetic field, B	2 T	2 T	Sec. III-A
Trigger cut	p_T(pair) > 200 GeV	p_T(pair) > 200 GeV	Sec. III-B
Detector acceptance	|η| < 1.8	|η| < 1.8	Sec. III-A
Beam-spot spread	4.5 cm	4.5 cm	Sec. III-A
Timing requirement	25 ns	25 ns	Sec. IV, after Eq.12

Additional background-related benchmark values:

Quantity	Value	Paper ref.
Beam-pipe conversion radius	2.35 cm (23.5 mm)	arXiv:1803.00844 Sec.2.1
Photon p_T cut	p_T(γ) > 0.5 GeV	Sec. III-D
Missing-energy (MET) cut	MET > 200 GeV	Sec. III-D
Professor-prescribed conversion weight	1% (0.01), applied as event weight	Project instruction
Z+γ+jet matched cross section (our run)	0.2310512 pb - OUR OWN value, not paper-quoted	N/A
EW quirk matched cross section (our run)	0.0002337 pb - OUR OWN value, not paper-quoted	N/A

4. MadGraph and Pythia8 Event Flow

- Colored quirk: process p p > t1 t1~ (+1 jet, MLM-matched), custom UFO model. Pythia8's RHadrons module (normally for SUSY particles) is reused to hadronize the quirk, exactly as the paper describes.
- EW quirk: process p p > qe+ qe- (+1 jet), a genuine Drell-Yan process — confirmed directly from the process card.
- Z+γ+jet background: generated the same way, with the Z forced to decay to neutrinos.

Verified fact: R-hadrons do not always keep the quirk's fractional charge. Some combine with a light quark to become electrically neutral, and a neutral particle leaves no signal in a silicon tracker. This is checked directly using Pythia8's own particle database (see Section 15).

5. Quirk Trajectory Implementation

The quirk pair's motion is calculated by numerically solving the equation of motion for each quirk (4th-order Runge-Kutta method, file: 03_trajectory_sim/quirk_trajectory.py). The force has two parts: a constant-strength pulling force from the string connecting the two quirks, and the ordinary magnetic (Lorentz) force from the 2 T field. The two quirks are evolved together, step by step, starting from the collision point.

6. Equations Implemented in the Code

Only equations that are genuinely implemented or directly used for validation are listed.

Equation 3 — String force and quirk motion

d/dt(mγv) = −Λ² [√(1−v⊥²)/v■ · v■ + v■/√(1−v⊥²) · v⊥] + F_ext

Symbol	Meaning	Unit
m	Quirk mass	GeV
v, γ	Quirk velocity, Lorentz factor	dimensionless (c=1)
Λ	String tension scale (Λ² = tension)	GeV
v■, v⊥	Velocity parallel/perpendicular to the string	dimensionless
F_ext	External force (magnetic Lorentz force)	GeV²

Code location: 03_trajectory_sim/quirk_trajectory.py, function string_force(). Paper reference: Sec. II, Eq. 3.

Meaning in simple words: the string always pulls the quirks back together with the same strength, no matter how far apart they are — this is what makes them oscillate instead of flying apart.

Equation 4 — Exact one-dimensional solution (used for validation)

d_cm(t) = (m_Q/Λ²) [γ − √(1 + (Λ²t/m_Q − vγ)²)]

Symbol	Meaning	Unit
d_cm(t)	Distance from center of mass to one quirk	cm
t	Time	GeV⁻¹ (natural units)
v	Initial quirk speed	dimensionless

Code location: 03_trajectory_sim/validate_against_eq4.py. Paper reference: Sec. II, Eq. 4. Meaning: this is the exact, solvable answer for a simple back-to-back case with no magnetic field — used purely to check that the general numerical solver is correct.

Equations 10–11 — Plane-fit distance and tensor

d(n,x_a) = √[(1/(N−1)) Σ■ (n·x■)²] T_ij = (1/(N−1)) Σ■ x■■ x■■

Symbol	Meaning	Unit
x_a	Position of hit a, from the primary vertex	cm
n	Candidate plane normal (unit vector)	dimensionless
N	Number of hits in the candidate	count
T_ij	3x3 tensor built from all hit positions	cm²

Code location: 05_coplanarity_analysis/coplanarity_fit.py, function fit_plane(). Paper reference: Sec. III C, Eqs. 10–11.

Meaning: T_ij is solved like a small matrix problem. Its smallest result tells us how “thin” the best-fit plane is (Δs); the next result tells us how “wide” the hit pattern is (Δw). A real quirk signal should have a very small Δs.

Equation 12 — Efficiency factorization

ε = ε_trig × ε_fid × ε_reco

Symbol	Meaning
ε_trig	Fraction of events passing the trigger cut (p_T(pair) > 200 GeV)
ε_fid	Fraction reaching all 8 layers within |η|<1.8 AND within 25 ns
ε_reco	Fraction of fiducial events passing the plane-fit signal-region cuts

Code location: 05_coplanarity_analysis/analyze_ew_quirk.py (cutflow structure). Paper reference: Sec. IV, Eq. 12. Meaning: the total chance of finding a quirk event is the product of three independent chances — passing the trigger, reaching the detector correctly, and being correctly reconstructed as a flat plane.

7. Magnetic-Field and Lorentz-Force Treatment

A uniform 2 T magnetic field along the beam axis (z) is applied to both quirks using the standard Lorentz force, F = q(v × B), added on top of the string force in Equation 3. The electric charge q is supplied in units of the electron charge (e.g. +2/3, –1, +1) and converted internally using e = $\sqrt{(4\pi\alpha)} \approx 0.30282212$, exactly once, inside the integrator.

8. ATLAS Eight-Layer Detector Model

The detector is modeled as 8 concentric cylindrical layers (4 Pixel/IBL + 4 SCT), matching real ATLAS Run-2 geometry.

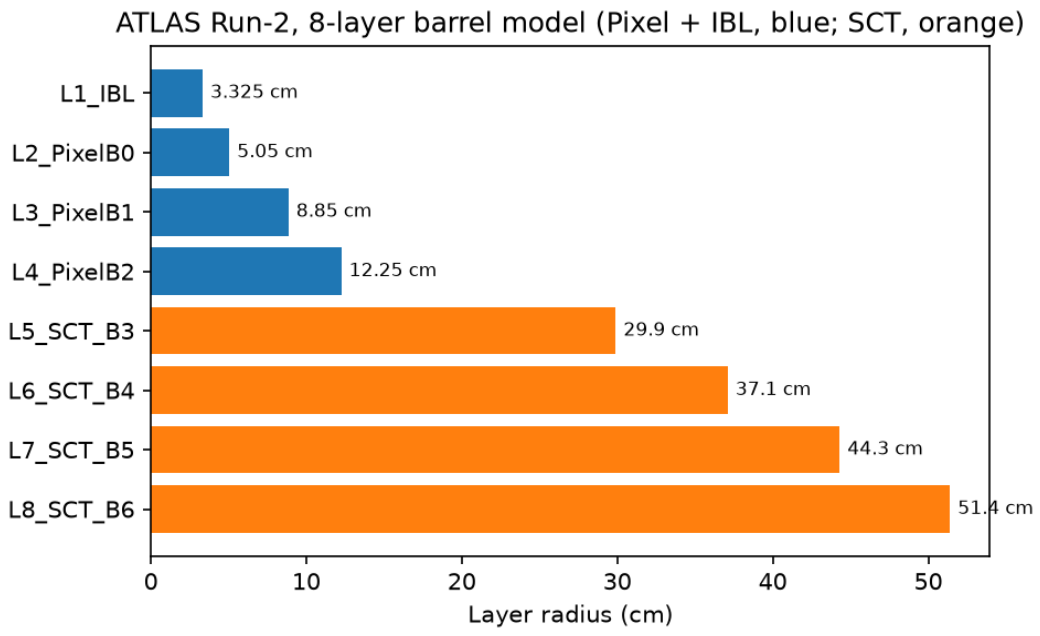


Figure 2. The 8 detector layers used in the simulation, with their radii.

Interpretation:

- The 4 inner layers (blue) are Pixel/IBL; the 4 outer layers (orange) are SCT strip layers.
- All 8 radii were checked directly against Table 1 of the ATLAS alignment paper (arXiv:2007.07624) and match exactly.

Layer	Radius (cm)	Type	r-ϕ pitch (μm)	z pitch (μm)
L1 (IBL)	3.325	Pixel	50	250
L2	5.05	Pixel	50	400
L3	8.85	Pixel	50	400
L4	12.25	Pixel	50	400
L5	29.9	SCT	80	n/a
L6	37.1	SCT	80	n/a
L7	44.3	SCT	80	n/a
L8	51.4	SCT	80	n/a

9. Pixel and SCT Detector Response

Pixel layers: the true hit position is rounded (“snapped”) to the nearest pixel, using the pixel size itself as the resolution — exactly as the paper prescribes. SCT layers: the true hit position is smeared with a Gaussian of width equal to twice the intrinsic accuracy (the paper's own “conservative doubling” choice). Hits from neighboring pixels are merged into one combined hit, using proper connected-cluster merging (fixed this session — see Section 15) so that a chain of three or more touching pixels always merges into a single cluster, and the merged position is correctly placed back on the true detector radius.

10. Coplanarity Plane-Fitting Algorithm

The plane-finding algorithm has three stages, following Sec. III C of the paper exactly:

- Seeding: find hit pairs in the two outermost layers (8 and 7) that are close in angle ($\Delta\phi<0.1$) and in z ($\Delta z<2$ cm), and combine them into a 4-hit starting plane.
- Iterative fitting: add one hit at a time from layers 6 down to 1, keeping only hits that do not make the plane's thickness or width grow by more than 3 times.
- Event selection: keep only the one candidate plane with $\Delta w < 1$ cm and the smallest Δs.

This was checked point-by-point against the paper's text with 12 separate unit tests (all in 05_coplanarity_analysis/coplanarity_fit.py) — 11 passed directly; one important issue (hits not measured from the true collision point) was found and is still open (see Section 15 and Section 18).

11. Trigger, Fiducial and Signal-Region Cuts

Cut	Requirement	Paper ref.
Trigger	p_T(quirk pair) > 200 GeV	Sec. III-B
Fiducial - geometric	Both quirks cross all 8 layers within $\|\eta\|<1.8$	Sec. IV
Fiducial - timing	Last layer crossing within 25 ns	Sec. IV, after Eq.12
Signal region - plane found	At least one reconstructed plane	Sec. IV, cut 1
Signal region - hit count	7 of 8 layers with >=2 hits; remaining layer with >=1	Sec. IV, cut 2

Cut	Requirement	Paper ref.
Signal region - shape	$\Delta w < 1.0$ cm and $\Delta s < 0.01$ cm	Sec. IV, cut 3

12. Converted-Photon Background

A photon from the Z+γ+jet process can convert into an electron-positron pair inside the beam pipe. This pair can sometimes fake a flat-plane signal. The conversion point is calculated at the true beam-pipe radius (2.35 cm), not at the primary vertex, and the pair is passed through the exact same detector and plane-fitting code used for the real signal (no confining string, ordinary charges –1/+1).

13. Professor-Prescribed 1% Conversion Weighting

Rather than randomly keeping only 1% of photons (which wastes 99% of the simulated events), every photon is converted and analyzed, and the known ~1% real-world conversion probability is applied afterward as an event weight. This is the professor-prescribed approach and is statistically more efficient than random sub-sampling.

14. Completed Validation Tests

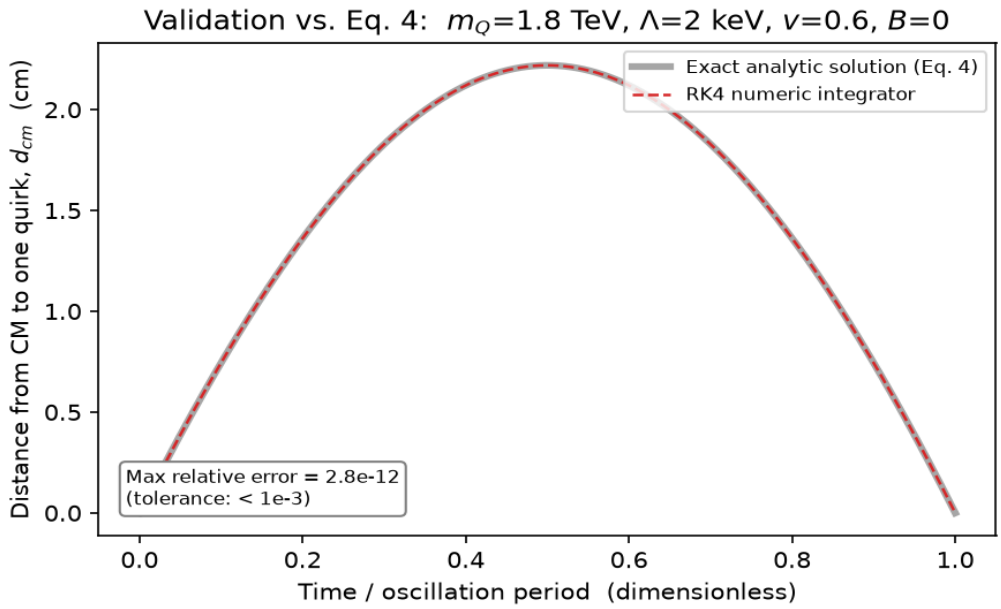


Figure 3. Numerical trajectory solver compared to the exact analytic solution (Eq. 4).

Interpretation:

- The red dashed (numerical) and grey solid (exact) curves lie on top of each other.
- Maximum difference is 2.8×10■¹² — far smaller than the 10■³ tolerance needed.

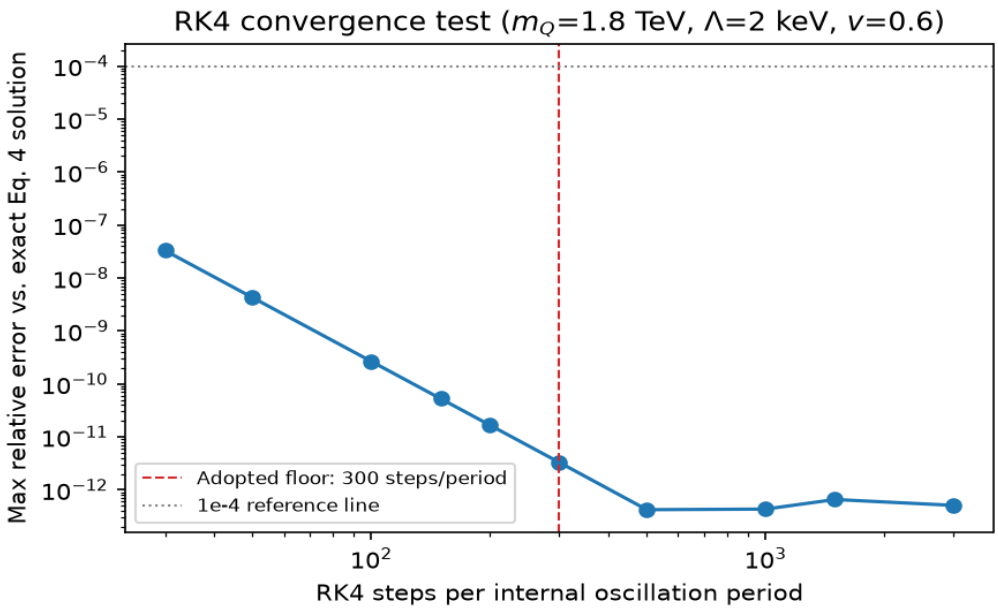


Figure 4. Accuracy of the numerical solver as the step size becomes finer.

Interpretation:

- Error drops steadily as more steps per period are used, confirming the solver behaves correctly.
- Even far below our chosen safety floor (300 steps/period, red line), the error is already 10,000 times smaller than required — so 300 is a safe, well-tested choice, not a guess.

Test	Result	Pass/Fail
Eq. 4 exact solution (3 benchmark points)	Max error 2.8e-12 to 9.3e-13	PASS
Solver convergence (step-size scan)	Error shrinks ~16x per halving, as expected	PASS
Charged-particle bending in B-field	Matches formula to 1.9e-9	PASS
Opposite charges bend opposite ways	Confirmed directly (Figure 5)	PASS
Pixel snapping, cluster merging, wraparound	10 targeted unit tests, all passed	PASS

15. Important Bugs Found and Corrected

Every item below was found using the actual project code and data, not guessed, and each is committed to the project's git history.

15.1 Reversed electric charge for the EW quirk (critical)

The two EW quirk particles were assigned the wrong sign of electric charge, bending every simulated EW-quirk trajectory in the wrong direction for the whole lifetime of the project. Found by checking the MadGraph particle-definition file directly.

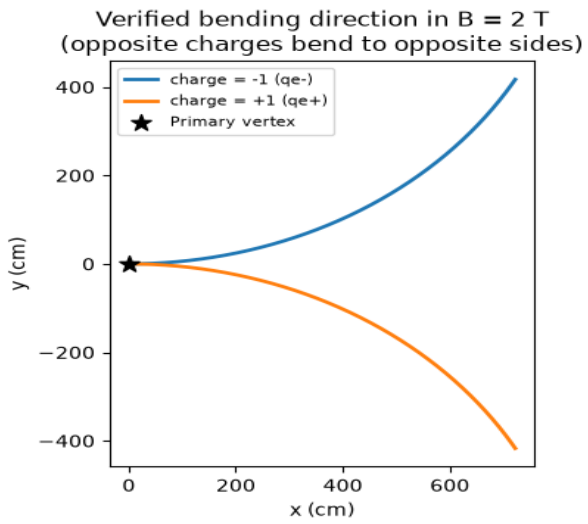


Figure 5. Verified: a charge of -1 bends one way and +1 bends the other way in the 2 T field.

Fixed in commit 4501105.

15.2 Colored-quirk R-hadrons can be electrically neutral

About 70% of colored quirk-hadron pairs include a neutral particle, which leaves no tracker signal at all. This was not being checked — every pair was treated as usable.

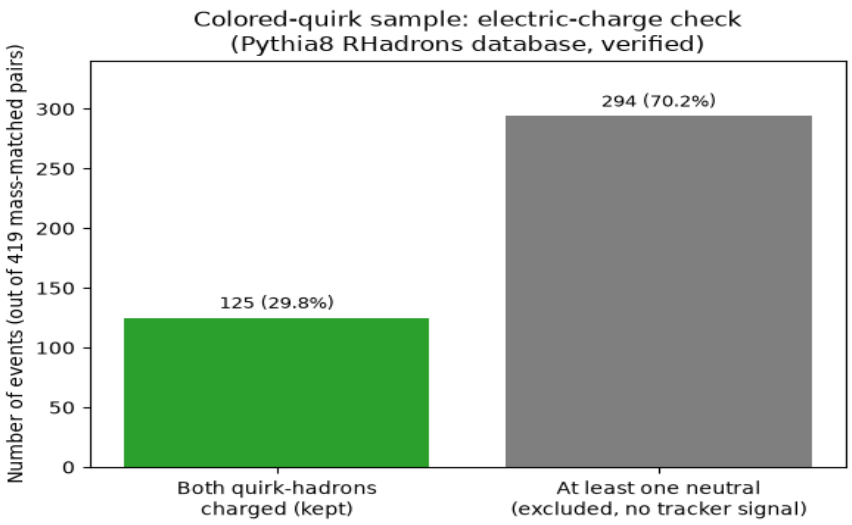


Figure 6. Charge check on the real colored-quirk sample, using Pythia8's own particle database.

Interpretation:

- Only 125 of 419 pairs (29.8%) have both particles charged and usable.
- This 29.8% closely matches the paper's own quoted “~30%” figure — strong evidence the physics is being modeled correctly, and that this check was genuinely missing before.

Fixed in commit ef9fe0a.

15.3 Missing 25 ns timing cut

The paper explicitly includes a 25 ns timing requirement inside the fiducial efficiency, but the code only recorded timing as information, without ever rejecting late events. Fixed in commit 4501105.

15.4 Trajectory integration step size (“step-budget”) bug

An earlier fixed cap on the number of integration steps caused 77 of 89 (86.5%) real EW-quirk events to be simulated with too few steps per oscillation — an accuracy risk. The cap has been replaced with logic that always targets the scientifically validated minimum resolution (300 steps per oscillation, see Figure 4) directly. Fixed in commit e12f3a7.

15.5 Vertex-offset bug in the plane fit (found, not yet fixed)

The paper assumes hit positions are measured from the true collision point. The code currently measures them from a fixed detector origin instead, without correcting for the true (smeared) collision point.

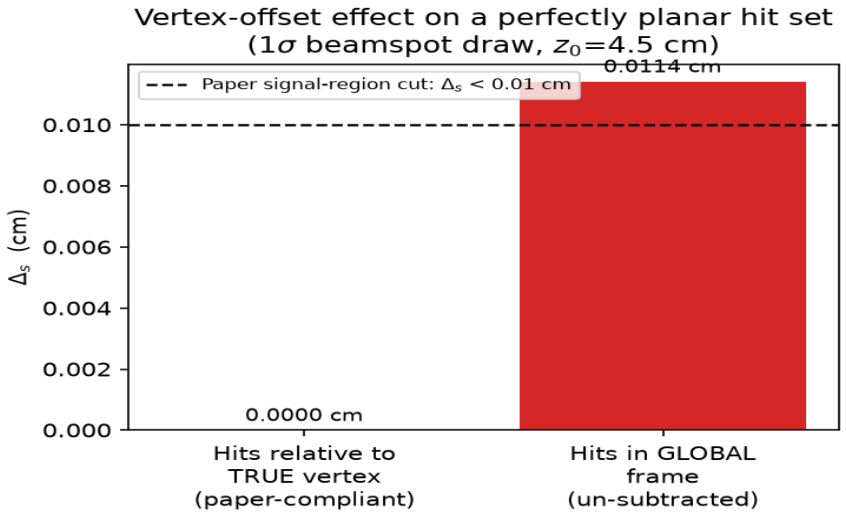


Figure 7. A perfectly flat hit set gets an artificial “thickness” if the true vertex position is not subtracted.

Interpretation:

- With the correct vertex subtraction, thickness is exactly zero, as it should be for a perfectly flat set of hits.
- Without it, for a typical beam-spot spread, the same perfectly flat hits appear to have a thickness of 0.0114 cm — above the paper's own signal-region cut of 0.01 cm.
- This is a likely cause of lower-than-expected reconstruction efficiency. Not yet fixed — see Section 18.

15.6 Other smaller fixes

- Pixel-cluster merging was not transitive (a chain of 3+ touching pixels did not always merge into one hit) — fixed.
- Signal-region cuts were being checked against the wrong candidate plane in some cases — fixed.
- The photon p_T cut and PDG-code checks were verified explicitly rather than assumed — both were already correct (zero events affected), but are now checked in code, not just assumed.

16. Verified Numerical Results

No final, corrected signal-region efficiency table exists yet for either benchmark (see Sections 17–18). The numbers below are the verified, current facts about the samples themselves.

Quantity	Value	Status
Colored-quirk generated pairs (mass-matched)	419	From MadGraph/Pythia8
Colored-quirk pairs with both particles charged	125 (29.8%)	Verified via Pythia8
EW-quirk generated pairs (matched)	511	From MadGraph/Pythia8
EW-quirk pairs passing trigger (p_T>200 GeV)	89	Truth-level, exact
Eq. 4 validation, max relative error	2.8e-12	PASS (tolerance 1e-3)

17. Current Work in Progress

A full, corrected EW-quirk run (with the reversed-charge fix, the 25 ns timing cut, and the step-budget fix all applied together) is running now.

Item	Status as of this report
Events complete	70 of 89 (as of the last check before this report)
Process health	Running normally, no errors
Estimated time remaining	Well under one hour
Final output file	ew_quirk_800gev_4kev_cutflow_v2_CHARGE_TIMING_FIXED.csv - NOT YET CREATED

This run is NOT complete and its results must not be treated as final or quoted as an efficiency number until all 89 events finish and the output file is validated.

18. Completed versus Remaining Work

Area	Status
Signal generation (both benchmarks)	DONE
Trajectory physics and validation	DONE
Detector geometry and response	DONE
Coplanarity algorithm mechanics	DONE (one open bug, see 15.5)
Vertex-offset correction in the plane fit	NOT YET DONE
Colored-quirk full corrected cutflow	NOT STARTED (old result is OBSOLETE)
EW-quirk full corrected cutflow	IN PROGRESS
Pile-up background simulation	NOT STARTED
Figure 4 (single clean event plot)	NOT REBUILT - old version is OBSOLETE
Figure 5 (full signal + background comparison)	NOT REBUILT - old version is OBSOLETE
Table I (full efficiency grid)	NOT STARTED beyond the two single points above
Figure 6 (exclusion/discovery contours)	NOT STARTED; needs external data we do not have

19. Important Limitations and Remaining Work

- The vertex-offset bug (Section 15.5) affects every plane-fit result produced so far, including the EW run currently in progress. It should be fixed before any efficiency number is treated as final.
- The colored-quirk benchmark has no working, up-to-date analysis script — it must be rebuilt using the same corrected structure as the EW-quirk script.
- Figures 4 and 5 from the paper cannot yet be honestly reproduced: doing so needs the vertex-offset fix, a corrected colored-quirk run, and a pile-up background that does not exist yet. Old, pre-fix versions of these figures exist in the project but are obsolete and are deliberately excluded from this report.
- Figure 6 needs external HSCP/monojet exclusion-curve data that the paper says came from private communication — this cannot be reproduced from the paper alone.
- The SCT detector's z-direction resolution (580 μm) could not be confirmed from the two ATLAS references available in this project; it is very likely correct but its exact original source is still unverified.

Appendix: File and Commit Reference

Item	Path / Commit
Project root	/home/rafia/coplanarity_study
Charge-sign + timing-cut fix	commit 4501105
Charge filtering (colored quirk)	commit ef9fe0a
Exact-PDG check (EW quirk)	commit b0361a8
Step-budget fix	commit e12f3a7
Photon p_T cut added	commit 5fa5506
Trajectory/detector correction set	commit 5abe83f
EW run output (when complete)	05_coplanarity_analysis/ew_quirk_800gev_4kev_cutflow_v2_CHARGE_TIMING_FIXED.csv
Obsolete EW file (do not use)	05_coplanarity_analysis/ew_quirk_800gev_4kev_cutflow.csv
Obsolete colored-quirk file (do not use)	05_coplanarity_analysis/colored_quirk_1800gev_2kev_cutflow_v2_FIXED.csv